Agentic AI

Super AI Engineer Level 2

Nutchanon Yongsatianchot (Nut)

CHA

10th ANIVERSARY

Workflows

- **Workflow**
- AI Agents
- Context Engineering
- How to fix your context
- Agent Summary

CHA

How can LLMs solve multi-step tasks?

Workflows are systems where LLMs and tools are orchestrated through **predefined code** paths.

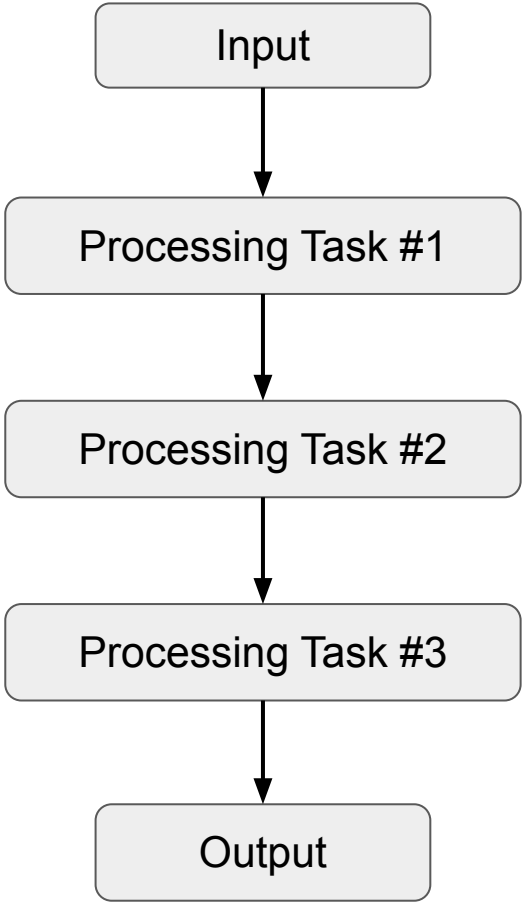
- Use when you can picture the exact flow or steps that AI should go through to solve a problem, or when your app's UX constrains users to a set of tasks

Agents are systems where **LLMs dynamically direct their own processes and tool usage**, maintaining control over how they accomplish tasks.

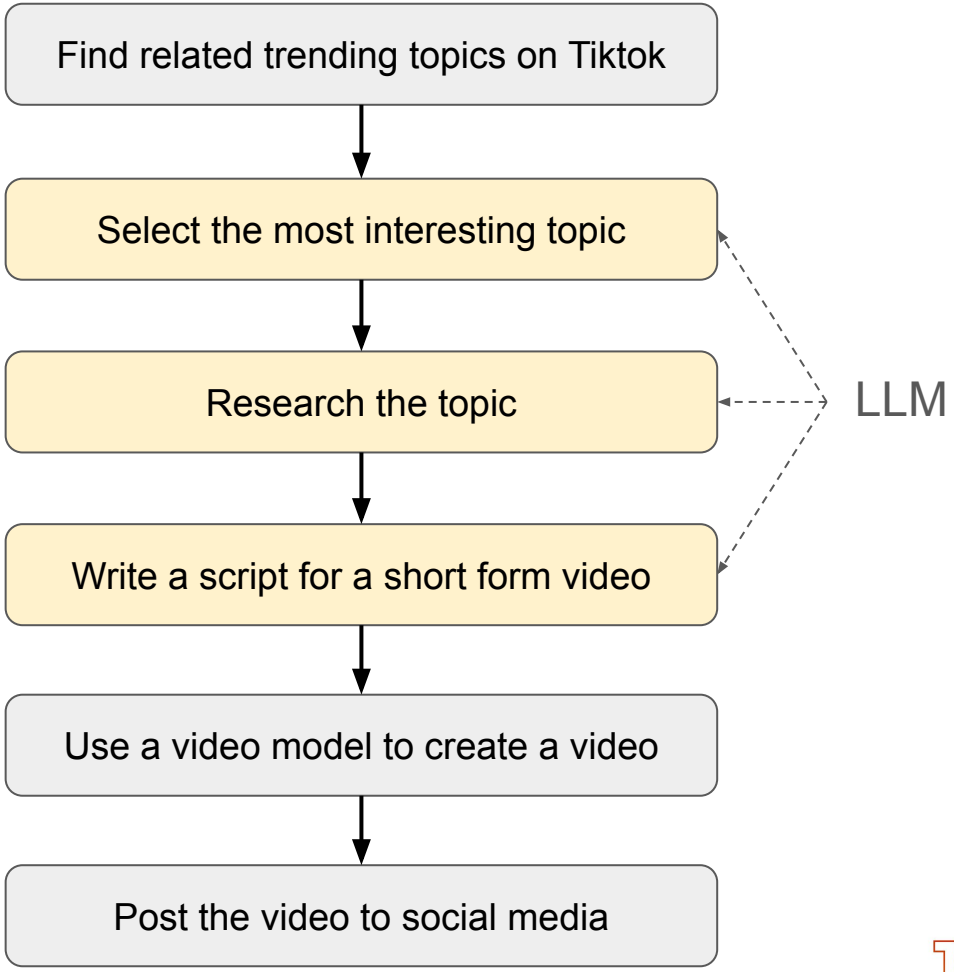
- Use when you're not sure exactly what task or task parameters you'll give to AI

Workflow: Prompt chaining

- Split large tasks into smaller, non-parallelizable subtasks
- Optionally do non-LLM processing between each task
- Keep AI focused on one aspect of the overall task



Workflow to automate the creation and posting of videos to social media



CHA

10th ANIVERSARY

Workflow #2: Stock News Analyst

[User: company name, e.g. "NVIDIA"]

|

▼

Step 1 (Web Search): "Search for latest news about company in the past week"

|

▼

Step 2 (LLM): "Summarize the top 3 news stories and classify each as positive, negative, or neutral"

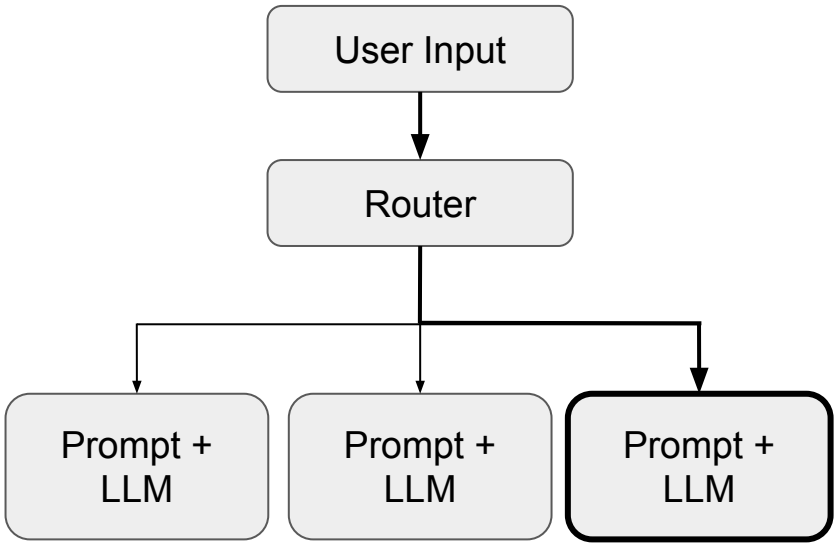
|

▼

Step 3 (LLM): "Based on the summary, write a 3-sentence investment outlook for a beginner investor"

Workflow: Routing

- User input goes to a router component first
- The router categorizes the request using an initial LLM call
- Based on the category, the input gets forwarded to one specific processing pipeline
- Each pipeline can have its own workflow, prompts, or tools optimized for that category



User input only sent to one

Router: "Customer Support Router"

[User: customer message]

▼

Classifier: "Classify this message as: billing, technical, or general"

│

├─ **billing** → "You are a billing specialist. Look up refund policy and help the customer..."

├─ **technical** → "You are a tech support agent. Troubleshoot step by step..."

└─ **general** → "You are a friendly FAQ bot. Answer briefly..."

Router: "Customer Support Router"

[User: homework question]

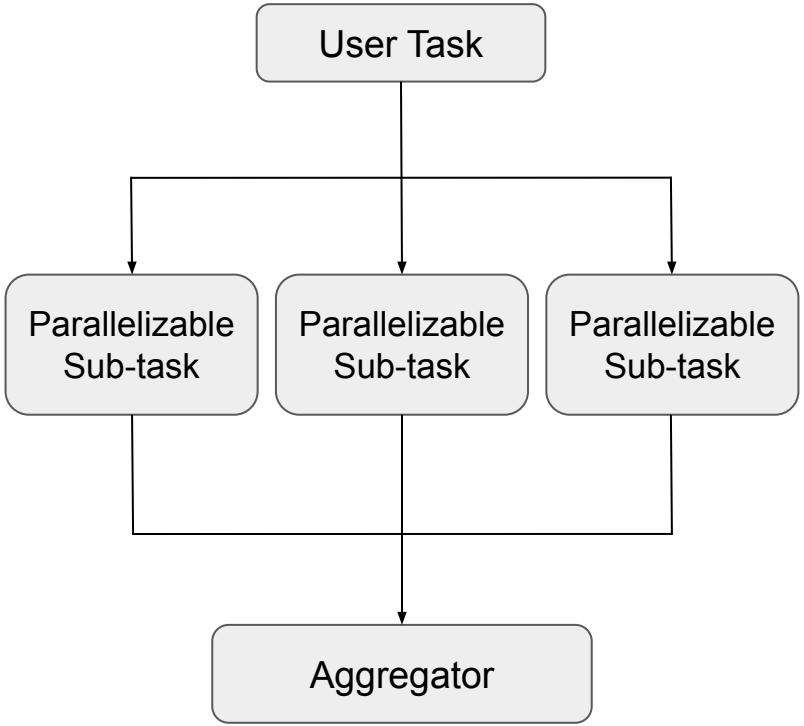
```
    |
    ▼
```

Classifier: "Classify this question as: math, coding, or essay"

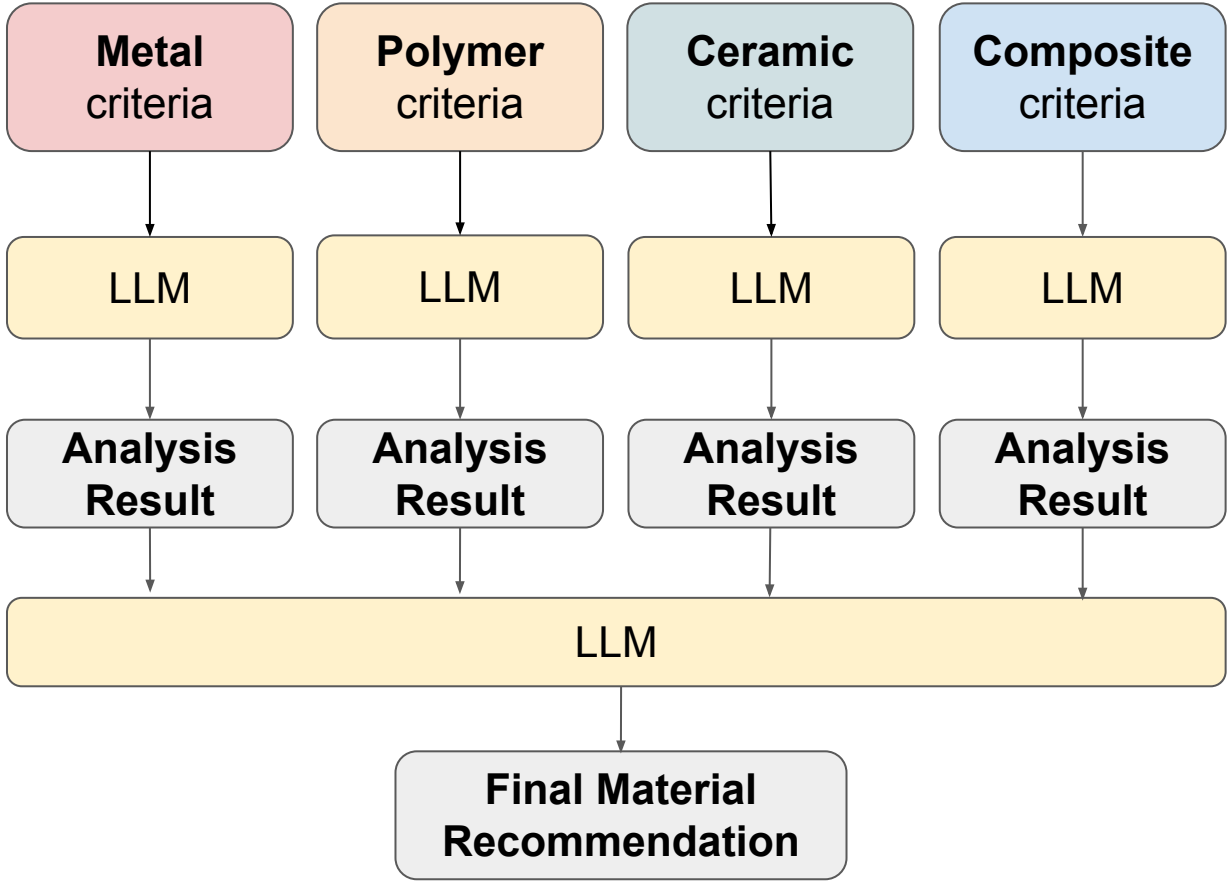
```
    |
    ├─ math   → (Code Interp) "Solve this step by step with Python, show the work"
    ├─ coding → (LLM) "Explain the concept first, then give code with comments"
    └─ essay  → (Web Search + LLM) "Search for key facts, then write an outline with sources"
```

Workflow: Parallelization

- **Split a single task into multiple sub-tasks**
- **Run the sub-tasks in parallel**
- **Aggregate the results together** - Combine the specialized analyses into a final decision
- **The parallelized sub-tasks don't need to be identical** - Each can have a specialized prompt, set of tools, or evaluation criteria



Prompt: Analyze the part in the attached image then decide if it would be best to make it out of a metal, polymer, ceramic, or composite.



CHA

Parallel 2: "Product Launch Review"

[User: product description, e.g. "A smart water bottle that tracks hydration"]

```
    |
    ├── Agent A (LLM): "List 3 strengths of this product idea"
    ├── Agent B (LLM): "List 3 weaknesses or risks"
    ├── Agent C (Web Search + LLM): "Search for competing products and list top 3 competitors"
    |
    ▼
```

Aggregator: "Combine the strengths, weaknesses, and competitor analysis into a 1-page product launch brief"

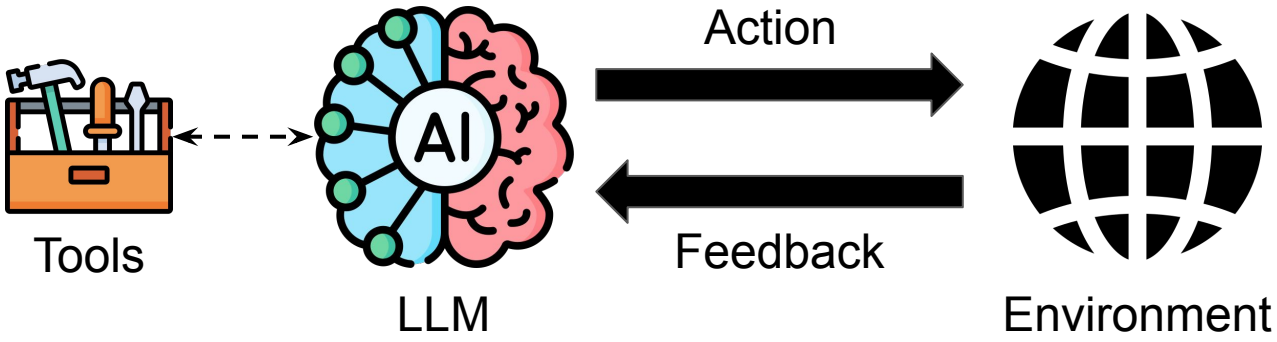
Agentic AI

- Workflow
- **Agentic AI**
- Context Engineering
- How to fix your context
- Agent Summary

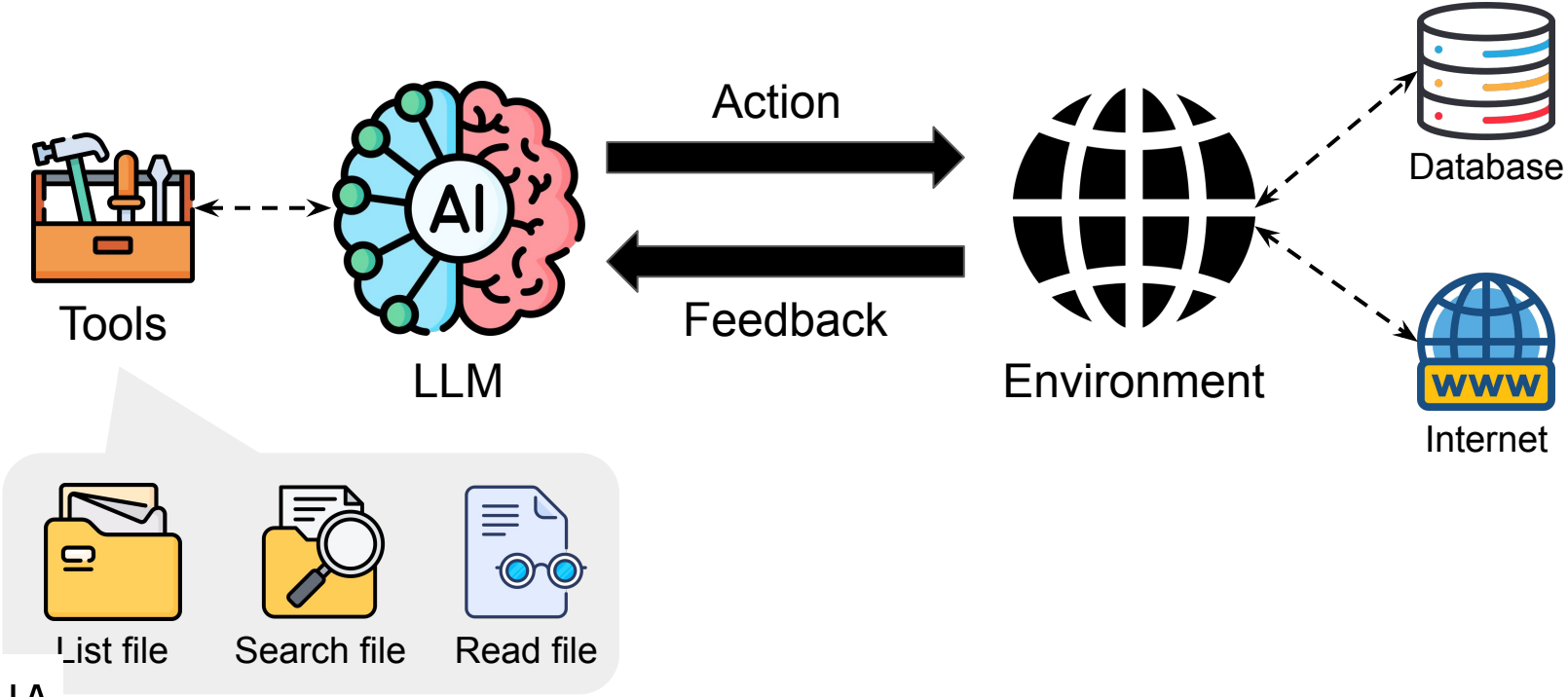
CHA

10th ANIVERSARY

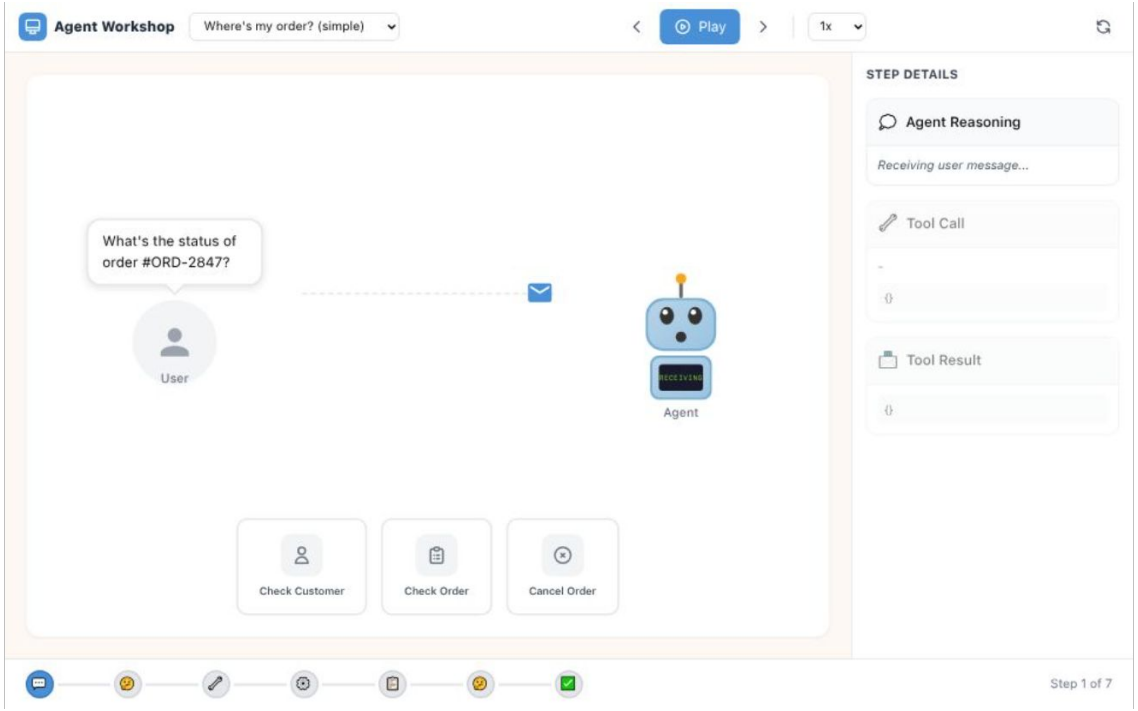
Agentic AI: LLM with tools in a loop



Agent Example



Example: Customer support



https://agent-workshop-iota.vercel.app/

Example: Agentic Search

Let’s LLM search information by itself!

CHA

10th ANIVERSARY

Example: Agentic Search

```
> Hello

● Hello! How can I help you today?

> check out the PRD in there. Tell me what you understand

● Search(pattern: "**/*PRD*")
  └ Found 1 file (ctrl+o to expand)

● Search(pattern: "**/*prd*")
  └ Found 0 files

● Read(Agent-Workshop-PRD.md)
  └ Read 557 lines

● Here's my understanding of the PRD:

  Agent Workshop - Summary

  What it is: An interactive, animated web app that visualizes how agentic LLMs work. It's an educational tool that show
s the complete flow from user query to agent response, making the "magic" of AI tool use tangible.

  Target audience: Students, developers new to AI, educators, and curious users.
```

CHA

10th ANNIVERSARY

Q: A 30-year-old patient is taking **Prozac** to treat their depression. They also carry WHIM syndrome and require a treatment for that condition as well. Is **Xolremdi** suitable for this patient, considering contraindications?

↓

get_indications: "Xolremdi"
get_contraindications: "Xolremdi"

↓

Indication: "…patients 12+ with WHIM syndrome…"
Contraindication: "…drugs highly dependent on CYP2D6 for clearance…" – FDA drug labels

↓

get_drug_interactions: "Prozac"

↓

"**Prozac** is a potent inhibitor of CYP2D6 pathway... **Prozac** is metabolized by CYP2D6…"
– FDA drug labels

↓

No, Xolremdi is not suitable for the patient. It is indicated for WHIM but contraindicated with drugs highly dependent on CYP2D6 for clearance. **Prozac** is metabolized by CYP2D6 and a potent inhibitor of this enzyme pathway. Therefore, there is potential for serious drug interactions.

CHA

10th ANIVERSARY

Example: Coding Agents

Agents are particularly effective for coding because:

- Code solutions are verifiable through automated tests;
- Agents can iterate on solutions using test results as feedback;
- The problem space is well-defined and structured; and
- Output quality can be measured objectively.

Agentic Walkthrough

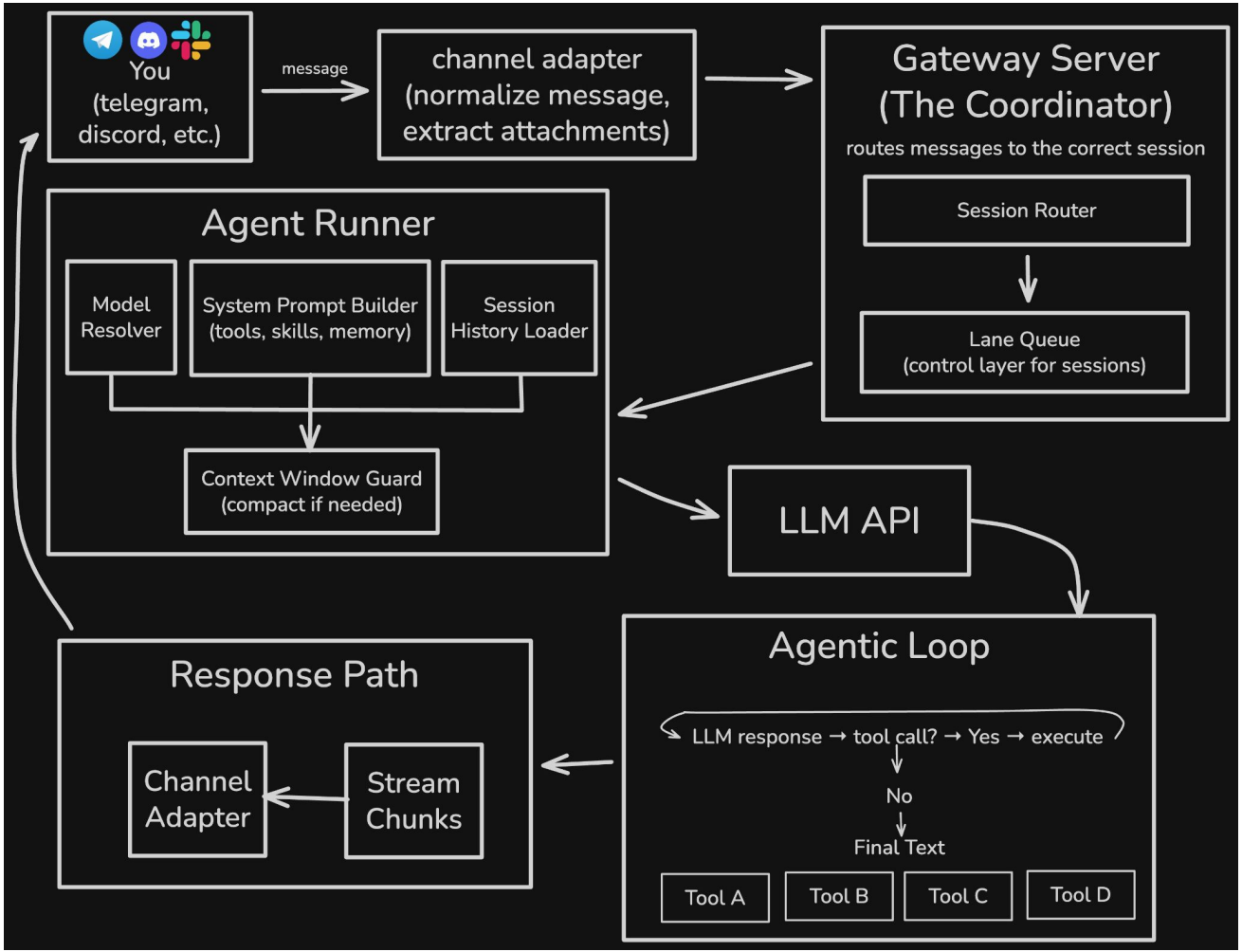
https://colab.research.google.com/drive/1VKfrxX5qW1_KAqQlyQ3PvSX86HkkEwjw?usp=sharing

Example Coding Agent

PI Coding agent →

```
You are an expert coding assistant. You help users with coding tasks by
reading files, executing commands, editing code, and writing new files.

Available tools:
- read: Read file contents
- bash: Execute bash commands
- edit: Make surgical edits to files
- write: Create or overwrite files

Guidelines:
- Use bash for file operations like ls, grep, find
- Use read to examine files before editing
- Use edit for precise changes (old text must match exactly)
- Use write only for new files or complete rewrites
- When summarizing your actions, output plain text directly - do NOT use
cat or bash to display what you did
- Be concise in your responses
- Show file paths clearly when working with files

Documentation:
- Your own documentation (including custom model setup and theme
creation) is at: /path/to/README.md
- Read it when users ask about features, configuration, or setup, and
especially if the user asks you to add a custom model or provider, or
create a custom theme.
```

https://github.com/badlogic/pi-mono/tree/main/packages/coding



CHA

https://x.com/Hesamation/status/2017038553055

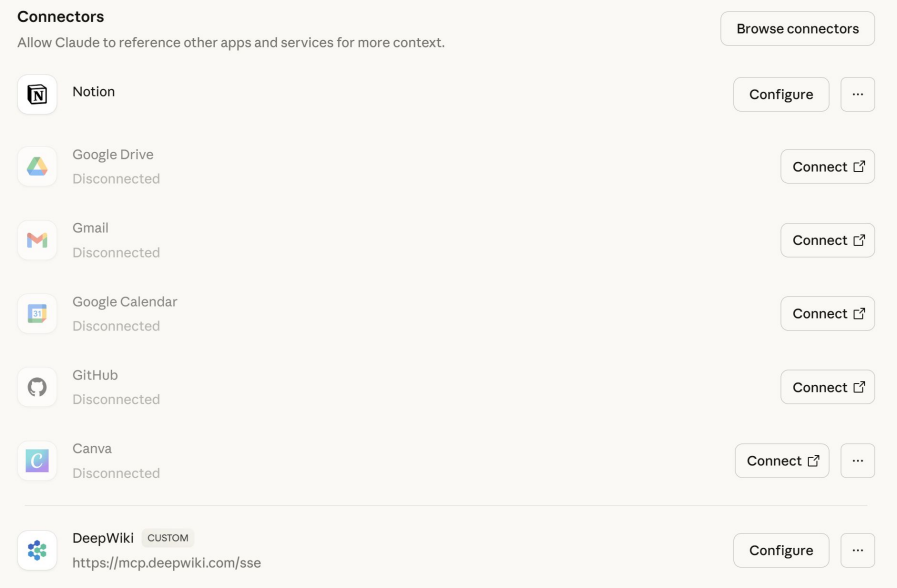
AIAT 10th ANIVERSARY

Example: Claude Code

Model Context Protocol (MCP)

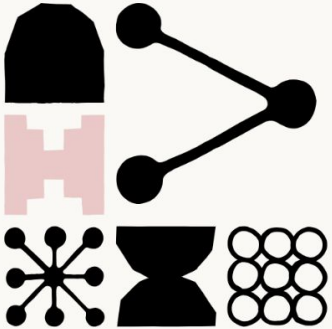
MCP provide tool schemas and functions already defined for you.

- You can simply add them to your agent.



Skills

Engineering at Anthropic



Equipping agents for the real world with Agent Skills

Published Oct 16, 2025

Claude is powerful, but real work requires procedural knowledge and organizational context. Introducing Agent Skills, a new way to build specialized agents using files and folders.

https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills

Context Engineering

- Workflow
- AI Agents
- **Context Engineering**
- How to fix your context
- Agent Summary

CHA

10th ANIVERSARY

… However, more tool uses → running out of context

- Context = input (prompts + tool results)
- Context = Working Memory
- **Context window is finite!**

CHA

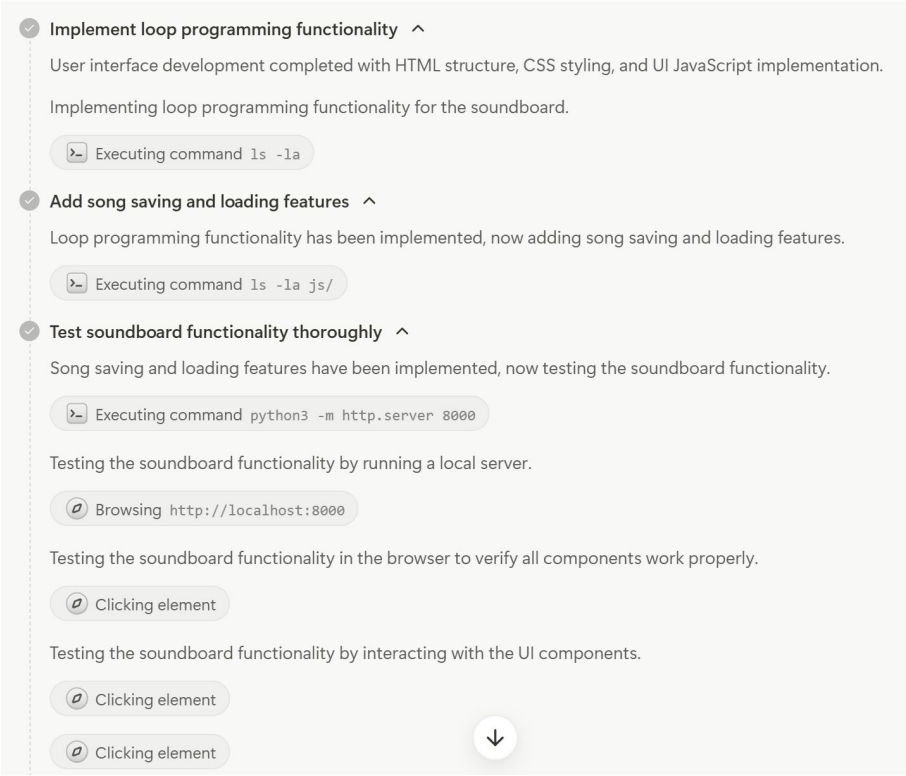
10th ANIVERSARY

Context grows with agents

Manus: Typical task in Manus requires around **50 tool calls**

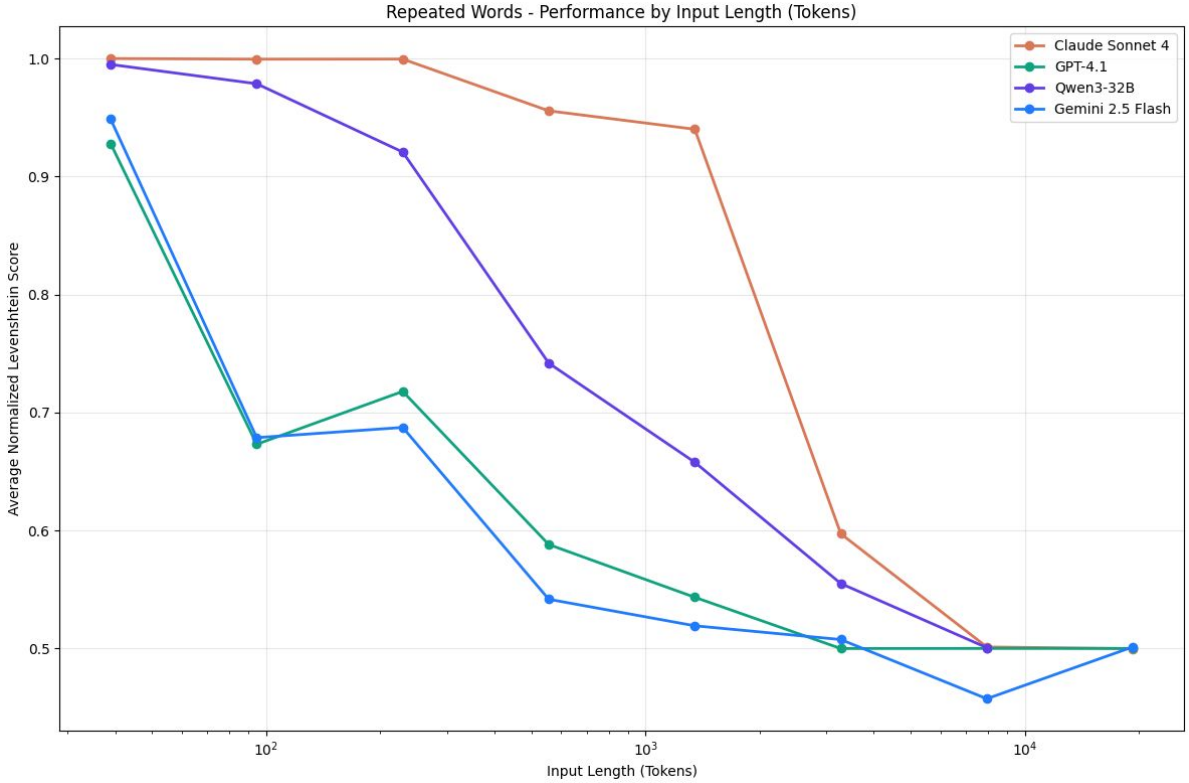
- Example

Anthropic: Production agents often engage in conversations spanning hundreds of turns



As context grows, cost and latency also increases.

Context Rot: As context grows, performances drop.



CHA

https://research.trychroma.com/cor

10th ANIVERSARY

Context Rot

100% context left

23% context left

Context Poisoning

Context Poisoning is when a hallucination or other error makes it into the context, where it is repeatedly referenced.

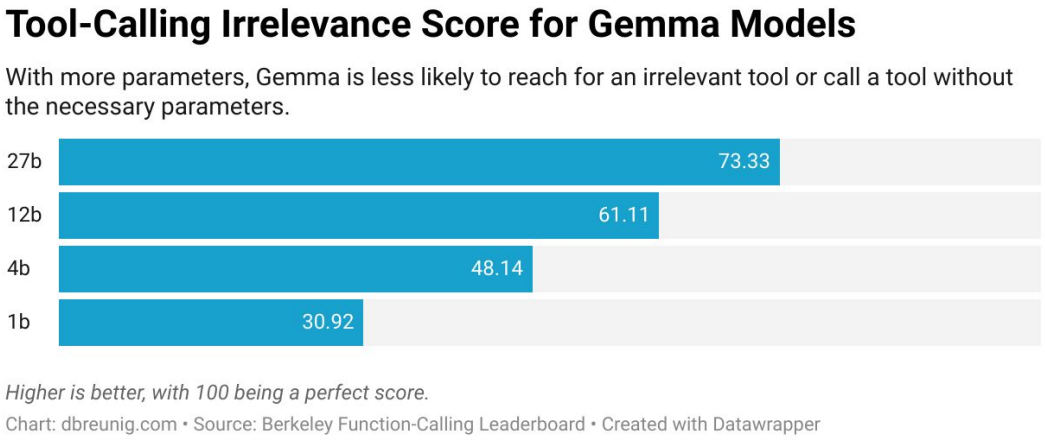
> An especially egregious form of this issue can take place with “context poisoning” – where many parts of the context (goals, summary) are “poisoned” with misinformation about the game state, which can often take a very long time to undo. As a result, the model can become fixated on achieving impossible or irrelevant goals.

- Gemini 2.5 pro plays pokemon

Context Confusion

Context Confusion is when superfluous content in the context is used by the model to generate a low-quality response.

- If you put something in the context the model has to pay attention to it.
- There can be too many tools



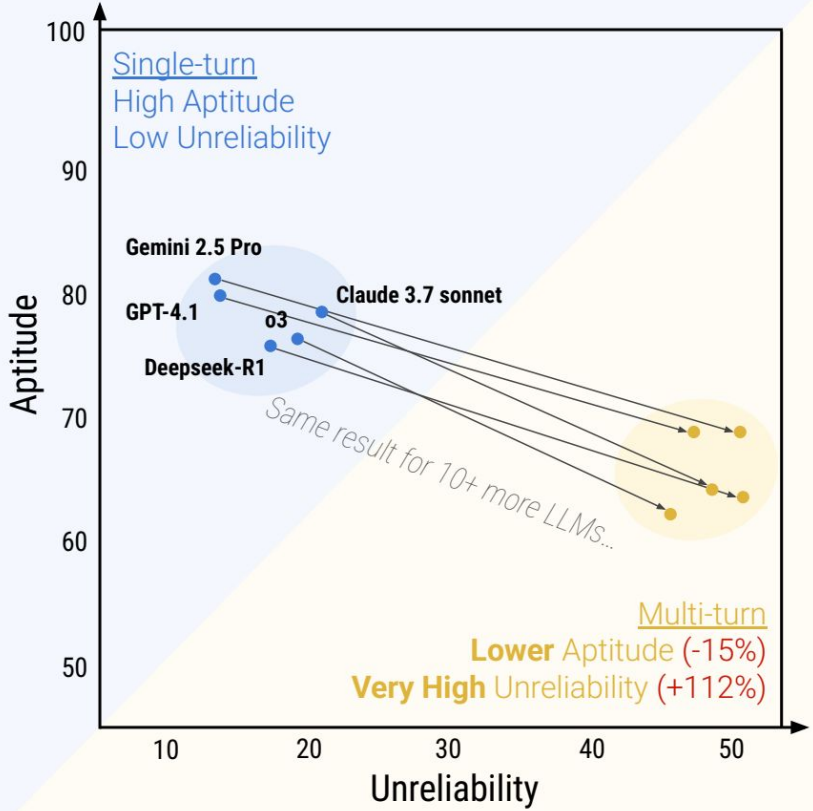
https://www.dbreunig.com/2025/06/22/how-contexts-fail-and-how-to-fix-th

LLMs get Lost in Conversation

Single-Turn Fully-Specified

User: Please generate X. I need [Requirement 1], [Requirement 2], also [Requirement 3].

Answer Attempt — LLM:
Sure thing!
```
def solution(x, y):
    [...]`
```



Multi-Turn Underspecified

User: I'm trying to implement X.

Clarification — LLM: Do you mean X' ?

User: No I want [Requirement 1].

Premature Answer Attempt — LLM:
Sure thing!
```
def function(x):
[...]
```

User: Well, I also need that [Requirement 3].

Incorrect Assumption — LLM:
Oh, in that case:
```
def function(x, y):
[...]
```

User: One more thing, can you include [Requirement 2]?

Bloated Answer — LLM:
Absolutely, here it is:
```
def function(y, x):
[...]
```


Laban, Philippe, et al. "Llms get lost in multi-turn conve

Fully-Specified Instruction (original)
Jay is making snowballs to prepare for a snowball fight with his sister. He can build 20 snowballs in an hour, but 2 melt every 15 minutes. How long will it take before he has 60 snowballs?

Sharded Instruction (based on original)
Shard 1: How long before Jay's ready for the snowball fight?
Shard 2: He's preparing for a snowball fight with his sister.
Shard 3: He can make 20 snowballs per hour.
Shard 4: He's trying to get to 60 total.
Shard 5: The problem is that 2 melt every 15 minutes.

The sharded prompts yielded dramatically worse results, with an average drop of 39%. And the team tested a range of models – OpenAI's vaunted o3's score dropped from 98.1 to 64.1.

> We find that LLMs often make assumptions in early turns and prematurely attempt to generate final solutions, on which they overly rely. In simpler terms, we discover that when LLMs take a wrong turn in a conversation, they get lost and do not recover.

Context Clash

Context Clash is when you accrue new information and tools in your context that conflicts with other information in the context.

This is a more problematic version of Context Confusion: the bad context here isn’t irrelevant, it directly conflicts with other information in the prompt.

Context Distraction

Context Distraction is when a context grows so long that the model **over-focuses on the context**, neglecting what it learned during training.

> While Gemini 2.5 Pro supports 1M+ token context, making effective use of it for agents presents a new research frontier. In this agentic setup, it was observed that as the context grew significantly beyond 100k tokens, the agent showed a tendency toward favoring repeating actions from its vast history rather than synthesizing novel plans. This phenomenon, albeit anecdotal, highlights an important distinction between long-context for retrieval and long-context for multi-step, generative reasoning.

https://www.dbreunig.com/2025/06/22/how-contexts-fail-and-how-to-fix-th

Context Distraction → In-context Learning

Don't Get Few-Shotted

Few-shot prompting is a common technique for improving LLM outputs. But in agent systems, it can backfire in subtle ways.

Language models are excellent mimics; they **imitate the pattern of behavior** in the context. If your context is full of similar past action-observation pairs, the model will tend to follow that pattern, even when it's no longer optimal.

This can be dangerous in tasks that involve repetitive decisions or actions. For example, when using Manus to help review a batch of 20 resumes, the agent often falls into a rhythm—repeating similar actions simply because that's what it sees in the context. This leads to drift, overgeneralization, or sometimes hallucination.

https://manus.im/blog/Context-Engineering-for-AI-Agents-Lessons-from-Building

Don't Get Few-Shotted

CONTEXT

...	
Action 1	Open http://example.com/1.html
Observation 1	Content saved to: /home/example_1.md
Action 2	Open http://example.com/2.html
Observation 2	Content saved to: /home/example_2.md
...	
Action k	Open http://example.com/k.html
Observation k	Content saved to: /home/example_k.md

CONTEXT

...	
Action 1	Open http://example.com/1.html
Observation 1	Content saved to: /home/example_1.md
Action 2	Open http://example.com/2.html
Observation 2	Content saved to: /home/example_2.md
...	
Action k	Open http://example.com/k.html
Observation k	This is already the k-th time visiting a page on ...

CHA

10th ANIVERSARY

In-context Learning (ICL)

- **ICL**: The model can learn from the **examples** you give it, and will detect and pick up any patterns in the text.
- ICL is highly unpredictable.

Circulation revenue has increased by 5% in Finland. // Positive

Panostaja did not disclose the purchase price. // Neutral

Paying off the national debt will be extremely painful. // Negative

The company anticipated its operating profit to improve. // _____

LM

Circulation revenue has increased by 5% in Finland. // Finance

They defeated … in the NFC Championship Game. // Sports

Apple … development of in-house chips. // Tech

The company anticipated its operating profit to improve. // _____

LM

https://arxiv.org/abs/2202

https://ai.stanford.edu/blog/understanding-inc

CHA

Chat History Effect

- In any long conversation, the biggest piece of context you have is your chat history.
- It's large and naturally repetitive, forming a strong pattern.
- This triggers **in-context learning** (ICL) and LLMs start to take the chat history as the narration example to follow.
- This effect is **self-reinforcing**, which makes it even worse.
- ICL is impossible for the model to resist if the pattern is strong enough.
- For these reasons, you **really really don't want** to play with a long chat history, and should trim it past maybe ~30 messages or so.

In-context learning is a fundamental LLM behavior. **It's extremely low-level, and powers almost everything in LLMs**.

Lost in the middle

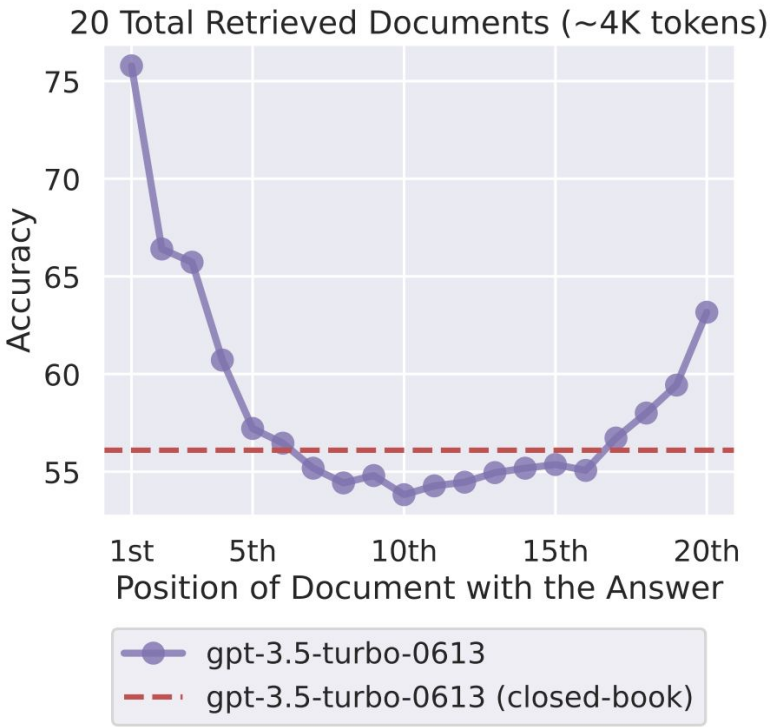
Input Context

```
Write a high-quality answer for the given question using only the provided search
results (some of which might be irrelevant).

Document [1](Title: Asian Americans in science and technology) Prize in physics for
discovery of the subatomic particle J/ψ. Subrahmanyan Chandrasekhar shared...
Document [2](Title: List of Nobel laureates in Physics) The first Nobel Prize in
Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...
Document [3](Title: Scientist) and pursued through a unique method, was essentially
in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics
Answer:
```

Desired Answer

```
Wilhelm Conrad Röntgen
```

Figure 2: Example of the multi-document question answering task, with an input context and the desired model answer. The document containing the answer is bolded within the input context here for clarity.



CHA

https://arxiv.org/abs/2307.0

Context fails: Issues as context grows

- **Context Poisoning** is when a hallucination or other error makes it into the context, where it is repeatedly referenced.
- **Context Confusion** is when superfluous content in the context is used by the model to generate a low-quality response.
- **Context Clash** is when you accrue new information and tools in your context that conflicts with other information in the context.
- **Context Distraction** is when a context grows so long that the model over-focuses on the context, neglecting what it learned during training.

“Context engineering” captures the challenges of the moment

tobi lutke
@tobi

I really like the term “context engineering” over prompt engineering.

It describes the core skill better: the art of providing all the context for the task to be plausibly solvable by the LLM.

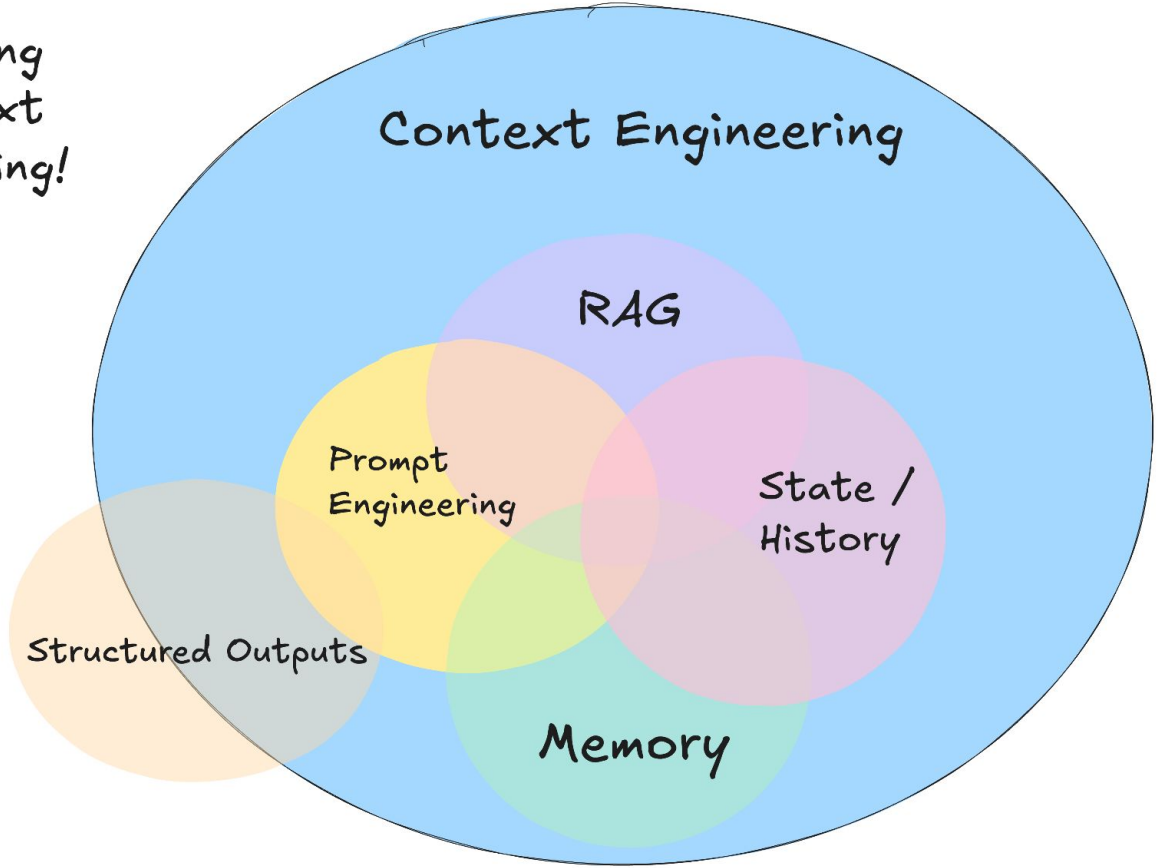
10:01 AM · Jun 19, 2025 · **1.9M** Views

Andrej Karpathy
@karpathy

+1 for "context engineering" over "prompt engineering".

People associate prompts with short task descriptions you'd give an LLM in your day-to-day use. When in every industrial-strength LLM app, context engineering is the delicate art and science of filling the context window with just the right information for the next step. Science

Everything is Context Engineering!



https://github.com/humanlayer/12-factor-agents/blob/main/content/factor-03-own-your-context-wir

How to fix your context

Adapted from https://x.com/RLanceMartin/status/1948441848978309358

- Workflow
- AI Agents
- Context Engineering
- **How to fix your context**
- Agent Summary

CHA

10th ANIVERSARY

How to fix your context

- Reduce Context
- Offload context
- Retrieve context
- Isolate context
- Cache Context

Reduce context

- Summarize agent message history
- Prune irrelevant parts of message history
- Summarize / prune tool call outputs
- **But, be careful of information loss!!**

Offload context: Use the File System as Context

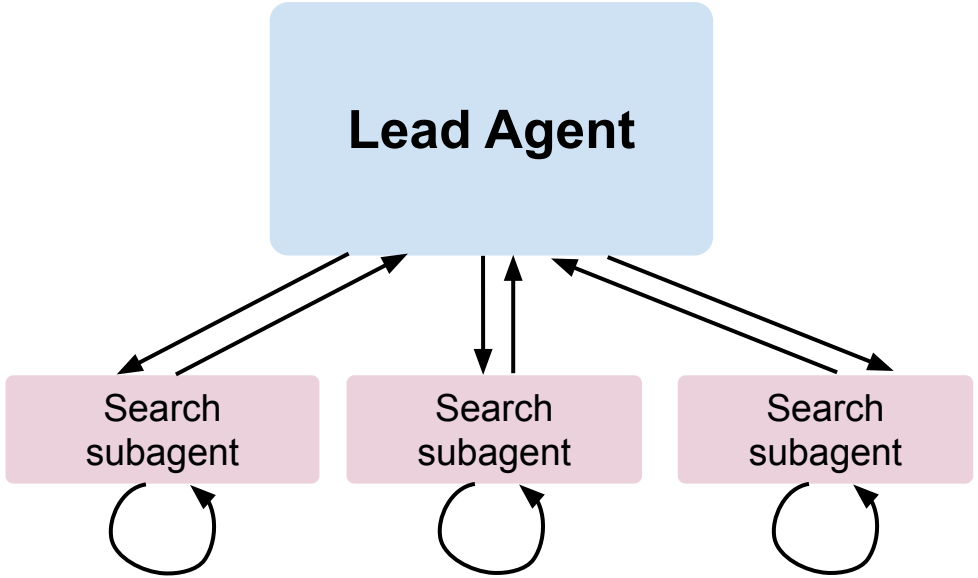
- Use file system for notes
- Use file system (e.g., todo.md) to plan/track progress.
- Use file system read/write token-heavy context.
- Use files for long-term memories.

Retrieve context

- Mix of retrieval methods + re-ranking.
- Systems to assemble retrievals into prompts.
- Retrieve relevant tools based upon tool descriptions.

Isolate Context: Multi-agent search solution

- Lead agent breaks down the question.
- One sub-agent per sub-question.
- Sub-agent returns only key information to the lead agent.
- Agents have tools to search and explore.



Cache Context

As context grows, you will have to pay more too!

- Cached input tokens for Claude-sonnet 10x cheaper!
- Cache agent instructions, tool descriptions to prefix.
- Add mutable context / recent observations to suffix.

Agent Summary

- Workflow
- AI Agents
- Context Engineering
- How to fix your context
- **Agent Summary**

CHA

10th ANIVERSARY

Don’t build agents for everything

Why not?

Agents scale **complex** and **valuable** tasks

Agents are **not** a drop-in upgrade for every use cases.

Checklist: “Should I build an agent”

Is the task complex enough?	No → Workflows Yes → Agents
Is the task valuable enough?	<$0.1 → Workflows >$1 → Agents
Are all parts of the task doable?	No → Reduce scope Yes → Agents
What is the cost of error/error discovery?	High -> Read-only/human-in-the-loop Low -> Agents

Why is coding a great agent use case

1. **Complexity**: design-doc to PR ✔
2. **Value**: $$$ ✔
3. **Viability**: Claude is great at coding ✔
4. **Cost of error**: Unit-tests + CI ✔

Summary

- **Tool use** allows LLM to interact with various applications.
- **Workflow** is a predetermined LLM system.
- **Agentic AI** is an LLM with tools in a loop.
- As agents work, context grows. **Managing context becomes critical.**
- Reliability becomes the critical aspect of agents.
- Evaluating agents is still tricky.
- **Human-in-the-loop**: Humans should be able to quickly verify agents' actions.